Assessing the potential impacts of weather changes on pest activities and establishment

Abstract

Early detection of invasive plant pests helps Plant Protection and Quarantine (PPQ) carry out its mission to safeguard agriculture and facilitate safe trade by assuring that our response to incursions is rapid and efficient, minimizing the costs of a response. The goal of this project was to develop a web app that delivers maps and other products that communicate the potential influence of recent (1981−2025) weather changes on the seasonal activities (phenology) and establishment of 18 major invasive insects for the contiguous United States. Project deliverables will inform strategic planning by PPQ and state agencies in terms of both when and where to conduct pest surveillance. For example, a trend towards the earlier appearance of adult insects may indicate that monitoring traps should be installed earlier in the season, whereas areas that are becoming increasingly suitable for pest establishment should be prioritized for surveillance.

Background

Recent weather trends may be promoting overwintering survival and increasing developmental rates of invasive insects, potentially allowing them to expand their range, emerge earlier, and attain higher densities over a longer growing season. The primary purpose of this project was to deliver predictions of the impacts of weather changes on the seasonal activities (phenology) and establishment risk of 18 major invasive insects that threaten agriculture in the United States to aid in early detection efforts. This project used a proven modeling system known as the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) to assess the potential impacts of recent weather trends (1981−2025) on the timing of pest activities (phenology) such as emergence, number of generations, and establishment for 18 major invasive species for the contiguous United States (Table 1).

Table 1. Invasive pests with completed DDRP models. Of the 18 species, two are included on the National Priority Pest (N=12) or Federal Program Pest (N=2) for 2025. Links to real-time forecasts, spreadsheet analyses, technical reports, and peer-reviewed publications associated with each species model are available at https://uspest.org/CAPS. References for technical reports or publications associated with each model, if available, are provided below.

Species	Common name	Established	National Priority Pest	Federal Program Pest
Anoplophora glabripennis	*Asian longhorned beetle*	Yes	Yes	Yes
Chilo suppressalis	*Asiatic rice borer*	No	Yes	No
Cryptoblabes gnidiella	*Christmas berry webworm*	No	Yes	No
Spodoptera litura	*Common or Cotton cutworm*	No	Yes	No
Agrilus planipennis	*Emerald ash borer*	Yes	No	Yes
Spodoptera littoralis	*Egyptian cottonworm*	No	No	No
Thaumatotibia leucotreta	*False codling moth*	No	Yes	No
Popillia japonica	*Japanese beetle*	Yes	No	Yes
Monochamus alternatus	*Japanese pine sawyer beetle*	No	No	No
Epiphyas postvittana	*Light brown apple moth*	Yes	No	Yes
Platypus quercivorus	*Oak ambrosia beetle*	No	Yes	No
Helicoverpa armigera	*Old world bollworm*	No	Yes	Yes

Dendrolimus pini	*Pine-tree lappet moth*	No	Yes	No
Autographa gamma	*Silver Y moth*	No	Yes	No
Neoleucinodes elegantalis	*Small tomato borer*	No	Yes	No
Lycorma delicatula	*Spotted lanternfly*	Yes	Yes	Yes
Eurygaster integriceps	*Sunn pest*	No	Yes	No
Tuta absoluta	*Tomato leaf miner*	No	Yes	No

Goals, Objectives, and Hypothesis

i. Goal

Our goal was to develop a web app that delivers maps and other products that communicate the potential influence of recent (1981−2025) weather changes on the phenology and establishment of 18 major invasive insects for the contiguous United States.

ii. Objectives

The project addressed the following three objectives:

1. Evaluate trends in predictions of selected phenological events (adult emergence and egg hatch) and climatic suitability over the 45-year study period.

2. Identify areas in the contiguous United States that consistently exhibit earlier pest activities, increased number of generations, and increased climatic suitability across the 18 species.

3. Deliver and promote interactive maps in a range of user-friendly, readily accessible formats via a web app at USPest.org (https://uspest.org/wea/), and through Extension programming and end user training.

iii. Hypothesis

We tested the hypothesis that species active earlier in the season and located in warmer regions where spring temperature velocity is lowest will show strongest sensitivity to climatic change and greatest advancement in response to warming (Guralnick et al. 2024). Temperatures in CONUS have risen by 2.5°F (1.4°C) since 1970, but there are substantial seasonal and regional variations in temperature trends (Vose et al. 2017, Marvel et al. 2023). We predicted that earlier phenological events and climatic suitability between 1981 and 2025 will be strongest in the northeastern and midwestern U.S. owing to particularly large increases (3 to > 5°F) in winter and spring temperatures in these regions since 1981 (Vose et al. 2017, Marvel et al. 2023).

Methods

All analyses and modeling were conducted using R versions 4.4 or 4.5 (R Development Core Team 2025). Raster data processing and calculations were performed using the R package *terra* v. 4.5 (Hijmans 2025).

All plots were produced using the R package *ggplot2* v. 3.5.2 (Wickham 2016).

i. Trend analyses for individual species

Five types of predictions produced by DDRP were retained for trend analyses: (1) dates (day of year) of first adult emergence, (2) dates of first egg hatch, (3) cold stress accumulation, (4) heat stress accumulation, and (5) the potential distribution, as measured by areas not excluded by cold and heat stresses. Modeling in DDRP used daily minimum and maximum temperature data for 1981 2025 at 4-km^2 resolution from the PRISM (Parameter-elevation Relationships on Independent Slopes Model (Daly et al. 1994, 2008) database (http://www.prism.oregonstate.edu).

We evaluated the direction and magnitude in change in predictions for 1981–2025 (entire 44 year period), 2001–2025 (recent period), and 1981–2000 (historical period) to provide insight into changes in phenology and climatic suitability at different temporal scales. Trends were assessed using a Mann-Kendall test and Theil-Sen regression, two ranked-based, non-parametric statistical methods that are widely used for trend analysis in time series of remote sensing data (Goossens and Berger 1986). Non-parametric methods were used because many atmospheric measurements, including temperature, strongly depart from a normal distribution. The Mann-Kendall test detects whether a monotonic upward or downward trend (increasing/decreasing) exists in time-series data and assess whether the trend is significant (p-value) (Mann 1945, Kendall 1975, Gilbert 1987). Theil-Sen regression is often used in conjunction with the Mann-Kendall test to provide a robust estimate of the slope by calculating the median of all possible slopes between pairs of data points, indicating units of change per time unit (e.g., 1 day/year for predicted dates of events).

The Mann-Kendall and Theil-Sen analyses were implemented in the R package *modifiedmk* v. 1.6 (Patakamuri 2021) and used predictions that were corrected for serial correlation (autocorrelation) using the approach of (von Storch 1995) and (Kulkarni and von Storch 1995). This correction for autocorrelation, also known as "prewhitening," is often applied to avoid violating the Mann-Kendall test's assumption of no autocorrelation in the time series, which may increase the likelihood of type 1 error (rejecting the null hypothesis when it might be true) (Kulkarni and von Storch 1995, Yue et al. 2002, Zhang and Zwiers 2004). The accumulation of degree-days and climate stress units in our models exhibits a seasonal cycle driven by temperature, which would increase autocorrelation in the time series (Coen et al. 2020).

A cumulative link model (CLM) was used to assess changes in predictions of the potential distribution because predictions were ordinal, with three categories corresponding to no climate stress exclusions, moderates stress exclusion, and severe stress exclusion (Barker et al. 2020). A lack of equal, interval-level spacing between these categories would make a linear regression analysis inappropriate. The CLM was implemented in the R package *ordinal* v. 2023.12.4.1 (Christensen 2025) and used the logit link function (proportional odds model). Positive coefficients in the CLM indicate an increased likelihood of establishment owing to reductions in survival-limiting climate stress, whereas negative coefficients indicate a lower likelihood of establishment owing to increasing stress.

ii. Comparative analyses

Outputs of trend analyses for the 18 species were combined to identify areas where pests consistently exhibit earlier pest activities, increases in the number of generations, and increases in climatic suitability. This involved calculating spatial descriptive statistics including the mean magnitude of change, variance in the magnitude of change, and the frequency of significant upward versus downward trends across species.

Outputs for each of the three time periods were analyzed separately.

Results